Predicting High-Resolution Root-Zone Soil Moisture Using an Analytical Relationship Between Satellite-Derived Soil Water Content and Soil Properties for Drought Monitoring

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***Abstract*—Root-zone soil moisture (RZSM) and surface soil moisture (SSM) are essential indicators for drought monitoring; however, studies on their effectiveness for drought monitoring is limited. For the first time, we investigated the use of a physically based model to predict RZSM at 50 cm depth and 100-m spatial resolution in the southern Great Plains (SGPs) of the United States for drought monitoring. The physically based model was primarily optimized using differential evolution algorithm and in situ observations (2015–2019) at site. Then, the optimized model was expanded to grid level (100-m resolution) with site-level parameters and covariates. Subsequently, we predicted RZSM from 2015 to 2019 at 100-m spatial resolution. The predicted RZSM showed acceptable accuracy during the optimization and validation periods, with root mean square error (RMSE) of 0.039 and 0.043 cm^3/cm^3, respectively, conforming to NASA's quality benchmark (RMSE = 0.06 cm^3/cm^3) for predicting soil moisture from satellite measurements. Additionally, we evaluated the accuracy of the standardized soil moisture index (SSMI), produced from the predicted RZSM and SMAP-HB SSM in capturing drought episodes with respect to the United States drought monitor (USDM) observations. The predicted SSMI-RZSM captured between 53% and 90% of the drought episodes, with an average of 71% between 2017 and 2019 across the validation stations while the SSMI-SMAP-HB only captured between 33% and 64%, with an average of 46%, indicating that SSMI-RZSM is more effective than SSMI-SMAP-HB at capturing drought episodes. We concluded that accurate and reliable measurements of root-zone field capacity and porosity are key to reliable RZSM predictions using the proposed approach. This study establishes a solid foundation for regional RZSM predictions and drought monitoring at higher spatial resolution.**

***Index Terms*—Drought, root-zone soil moisture (RZSM), soil properties, surface soil moisture (SSM).**

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I. Introduction

SOIL moisture directly or indirectly controls a manifold of land-atmosphere processes, including water, energy, and biogeochemical cycles [1]. Soil moisture plays a key role in agricultural production, water resource management, and climate change projections [2]. It is believed that soil moisture is the only environmental variable that effectively responds to climate forcings across varying climatic zones [3]. Considering the influence of soil moisture over land-atmosphere processes and agricultural systems, which directly impact living things, soil moisture studies are becoming increasingly important in the 21st century.

The root-zone soil moisture (RZSM) content refers to the water content in the unsaturated soil zone, which is available for plant roots [4], [5]. However, soil-rooting depth is determined by two core factors: soil property and crop/vegetation type; hence, there is no consensus on the minimum-maximum rooting depth [4]. Baldwin et al. [6] assumed 100 cm as the optimum rooting depth, and Romano et al. [3] calculated 82.5 cm as the effective rooting depth based on local soil, vegetation, and climate characteristics. Similarly, Rivieccio et al. [7] reported that the rooting depth can reach up to 150 cm under favorable soil conditions. However, White-Murillo et al. [2] reported that 75% of plant roots are generally within a depth of 40 cm.

As the world's population is increasing exponentially, with a prediction to reach 10 billion by 2050, ensuring food security is essential [2]. Maintaining global food security requires increasing agricultural productivity and mit-

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igating the impacts of climate change. The RZSM is a critical factor in agricultural production as it controls nutrient transport from soil to plants, which are key for crop health and optimum yield [7], [8], [9]. Therefore, adequate knowledge of RZSM can improve agricultural production through effective crop monitoring and agricultural water management.

Having realized the importance of soil moisture for optimum ecosystem and sustainable environment function, scientists employ different methods to measure it at different profile layers/depths and at various scales, ranging from point to areal footprint. Time domain reflectometry, cosmic ray neutron probes, and capacitive moisture sensors are used for point measurements [10]. Remote sensing methods for regional and global observations have been used for over the past four decades because of the shortcomings of point-based measurement for environmental monitoring at larger scales. Multiple microwave-satellite missions were launched in the last three decades for the surface soil moisture (SSM) measurement, notable to mention: SMOS, SMAP, and ESA-CCI missions. Many researchers have proposed new or improved methods for estimating SSM from optical/thermal infrared measurements and have validated the existing methods. Notable works to mention: [11], [12]. Estimating RZSM directly from satellite observations proved to be difficult. However, with recent developments in remote sensing technologies, different methods have been developed. Kasim et al. [4] broadly categorized the methods for estimating RZSM from remote sensing data into two groups: methods with knowledge of SSM and methods without knowledge of SSM. The methods of estimating RZSM with knowledge of SSM are mostly empirical, physics-based, and assimilation-based methods. On the other hand, the methods without knowledge of SSM are based on land surface temperature, energy fluxes, and low-frequency radar measurements. For details on these methods, their pros and cons, and future research directions, a reader is referred to Kasim et al. [4]. Although many satellite-based RZSM products [13], [14] and model-based/machine learning-based products [15], [16], [17], [18] are available, they suffer from certain uncertainties and coarse spatial resolutions. Xu et al. [19] evaluated the performance of the eight most widely used model-based global RZSM products for drought monitoring and other applications, including GLDAS NOAH, ERA-5, MERRA-2, NCEP R1, NCEP R2, JRA-55, SMAP level 4, and SMOS level 4. Although some products showed better performance than the others, none of the them exhibited optimal performance across all land cover types because of diverse sources of uncertainty including, methods of data assimilation and assimilated variables (e.g., radio frequency interference issue in SSM), forcing data, model structure, and parameterization.

RZSM is mainly required at high spatiotemporal resolutions, especially in agricultural production and water resource management applications. However, the satellite- and model-based products are produced at a high time step and coarse spatial resolution ranging from 9 to 25 km. Therefore, the utility of these products is limited. In addition to using aircraft missions (e.g., AirMOSS), empirical or physics-based methods are the only way to obtain a high spatiotemporal RZSM map. One such method is the soil moisture analytical relationship (hereinafter referred to as SMAR) infiltration model, which takes in SSM in conjunction with soil physical and hydraulic properties to estimate RZSM. The SMAR model was first developed by Manfreda et al. [20] and has been used to infer RZSM in the following studies: [6], [21], [22]. However, these previous works estimated RZSM at coarse spatial resolution ranging from 1 to 25 km, which is not suitable for agricultural and water resource management applications.

Generally, soil moisture is key to agricultural drought monitoring [23], and hence most indices are based on soil moisture measurements. Historically, drought episodes have been monitored using point-based/aboveground observations, which are often not available for long-term monitoring in many parts of the world. With advancements in Earth Observation systems, scientists use satellite-based/model-based SSM to compute drought indices at regional and global scales. Drought indices can be univariate, such as the standardized soil moisture index (SSMI), soil moisture anomaly percentage index (SMAPI) [24] or multivariate/integrated, such as the standardized precipitation, evapotranspiration and soil moisture index (SPESMI) [25] and US drought monitor (USDM). These types of drought indices differ in their ability to monitor drought types. Univariate indices are more effective in monitoring specific type of drought, such as agricultural drought (soil moisture deficit), while multivariate indices effectively monitor meteorological agricultural, and hydrological droughts [25].

Despite the importance of soil moisture in monitoring agricultural drought, its utility in depicting agricultural drought at a regional scale, considering surface heterogeneity, is limited because of the following reasons: 1) in situ observation networks are sparse and unevenly distributed, and therefore measurements cannot adequately capture spatial variations [26]; 2) model-simulated soil moisture is associated with biases because all land surface models have variable biases and performance from region to region [27]; and 3) satellite-based soil moisture products are limited to the top few centimeters of soils and their accuracy significantly varies from product to product and from region to region [26].

Consequently, this study is the first to investigate the feasibility of estimating RZSM at an unprecedented spatiotemporal resolution of 100 m/daily toward better agricultural drought monitoring at a regional scale. Specifically, this study aims at addressing three main research questions.

1) Can the proposed approach meet NASA's accuracy benchmark?
2) How to expand the RZSM model from site-level to grid-level?
3) Does the predicted RZSM dataset support drought monitoring?

The remainder of this article is organized as follows. Section II describes the methods and datasets, Section III presents the results, Section IV provides a discussion, and Section V concludes this article.

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II. Methods and Dataset

A. RZSM Prediction

SMAR is a physically based model that reliably predicts moisture content at rooting depths by inputting SSM and parameters estimated from soil physical properties. First developed by Manfreda et al. [20], SMAR model basically assumes that the infiltration of excess moisture from the surface soil layer to the rooting depth is a function of moisture content at the surface soil layer, rather than rainfall in the conventional water balance equation. Consequently, the moisture flux from the surface layer to the root-zone layers is trivial, except that it goes beyond the field capacity. Therefore, the infiltration flux from the surface soil layer to the rooting layer assumes to be instantaneous, within 24 h after rainfall. These analytical relationships between the two soil layer components can be expressed as follows:

$$S_2(ti) = \left[S_{wl} + (S_2(t_{i-1}) - S_{wl}) \exp^{-a(t_i - t_{i-1})}\right] + \left[(1 - S_{wl})\, b \times y(t_i)(t_i - t_{i-1})\right] \quad (1)$$

where $S_2(t_i)$ is the relative RZSM at time t_i and S_{wl} is the relative root-zone wilting level. a is normalized daily root-zone water loss coefficient and b is the diffusion rate coefficient, which is expressed in (2) and (3), respectively. $y(t_i)$ is a fraction of the moisture content infiltrating to the root-zone layer, at time, t_i

$$a = \frac{V_2}{(1 - S_{wl})\, n_2 \phi_2}. \quad (2)$$

The normalized daily root-zone water loss coefficient and the diffusion rate coefficient can be expressed as

$$b = \frac{n_1 \phi_1}{(1 - S_{wl})\, n_2 \phi_2} \quad (3)$$

where n_1 and n_2 denote the soil porosity at the surface and root-zone layers, respectively. ϕ_1 and ϕ_2 denote the soil thickness of the surface and root-zone layer, respectively. V_2 denotes daily leakage rate (cm day^{-1}).

Basically, the SMAR model has four parameters required to be determined: the relative field capacity of the surface layer, which is a function of $y(t_i)$, relative wilting level of the root-zone layer (S_{wl}), root-zone water loss (a), and diffusion coefficient (b). The relative field capacity of the surface layer illustrates the occurrence of infiltration from the surface to the root-zone layer. Precisely, when the surface moisture content surpasses the field capacity (the moisture threshold where drainage stops after heavy rainfall or irrigation events), the surplus water percolates downward to the root-zone within 24 h, illustrates the total absence of moisture at the rooting depth (the moisture threshold where plants cannot extract water). a represents combined water losses from evaporation, transpiration, and percolation at the rooting depth, while b is expressed as the ratio of the total possible water storage of the surface layer to the total possible water storage of the root-zone layer, controlling the redistribution of the surface moisture surplus to deeper soil layers [6], [20]. These parameters can be estimated with soil physical properties and evapotranspiration.

B. Model Optimization

Although very simple and straightforward, the differential evolution algorithm (DE) is a promising and heuristic approach for simple and complex global parameter optimization [28]. The stochastic nature of DE allows it to search the entire parameter space for candidate selection, and the greedy approach makes it find a solution faster than other evolutionary algorithms. Consequently, the DE was used to estimate optimal SMAR parameters fit to in situ RZSM measurements, which are the foundation for spatial parameter prediction, and subsequently RZSM across space. The goal of the minimization method, such as DE, is to optimize parameters based on the objective function, which makes the observed data the most probable and minimizes the prediction error. For details on DE, the following references are hereby referred to [29] and [30].

Assuming a Gaussian distribution, we used a negative log-likelihood (-LL) as an objective function/cost function, which maximizes the probability of the observed data (lower values equal better fit), and root mean square error (RMSE) and ubRMSE, which minimize prediction error, as fitness functions. The best candidate parameter was chosen using a greedy strategy on the basis of minimum error between the in situ and predicted RZSM using the SMAR function [31].

The log likelihood function was used as follows:

$$LL = \log L\left(\mu, \sigma^2; X\right) \quad (4)$$

and can be expressed as

$$LL = \frac{N}{2}\log(2\pi) - \frac{N}{2}\log\left(\sigma^2\right) - \frac{1}{2\sigma^2} SSR \quad (5)$$

where $N/2\log(2\pi)$ is normalization factor of the normal Gaussian distribution, $-N/2\log(\sigma^2)$ is variance contribution to the log likelihood, $-\mathrm{SSR}/2\sigma^2$ denotes the squared differences between the in situ and RZSM predictions as follows:

$$SSR = \sum_{i=1}^{n} \left(RZSM_{insitu,i} - RZSM_{predicted,i}\right)^2. \quad (6)$$

The SMAR model was initialized at sites with the prior parameters, including field capacity, wilting level, and porosity extracted from Rawls et al. [32]. Other initial inputs are site characteristics, including sand, silt, and clay fractions (at the surface and root-zone), and annual evapotranspiration (AET). Parameters were log-transformed prior to modeling, and physical bounds were set on the predictions such that they must not be less than the minimum prediction of the calibrated model or greater than the maximum prediction of the calibrated model.

C. Conversion of the SMAR Model From Site-Level to Grid-Level

Assuming a linear relationship exists between the predictors and the response variable, generalized linear models can offer straightforward and powerful solutions. However, in the event where a nonlinear relationship exists between the predictors and the response variable, such as in the case of soil hydraulic properties, the generalized additive model (GAM) is quite

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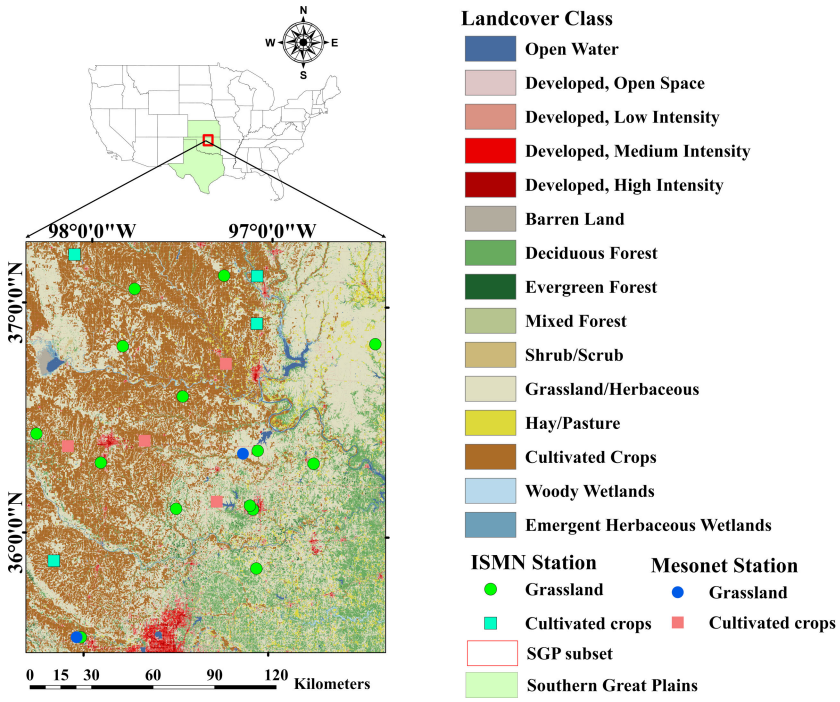


Fig. 1. SGPs and its subset. The ISMN and Mesonet stations represent the optimization and validation stations, respectively across the land cover types.

efficient and robust in capturing complex and nonlinear relationships by expressing a response variable as a sum of smooth functions of the predictors with an error term. GAM uses link and smooth functions. The latter model a nonlinear relationship between a response variable and predictors by smoothing the effect of each predictor to prevent overfitting, and the former link the expected value of the response variable to the additive predictor [33]. Additionally, GAMs are constrained by degrees of freedom, so over-fitting is less of an issue for them than more complex "black box" machine learning models. The generalized form of GAM is expressed as follows:

$$g\left(E\left[Y|X\right]\right) = \beta_0 + f_1\left(X_1\right) + f_2\left(X_2\right) + , \ldots, f_p\left(X_p\right) + \varepsilon \tag{7}$$

where Y is a response variable. X_1, X_2,..., X_p are predictors, ε is an error term, $g(.)$ is a link function (e.g., identity for Gaussian), $f_j(.)$ is a smooth function (e.g., thin-plate regression spline), β_0 is intercept.

We used the identity link function and thin-plate regression spline as a smoothing function on the assumption of Gaussian, which is easy to use with GAM and computationally efficient for a reasonable dataset size. To convert the SMAR model from site-level to grid-level using GAM, the soil properties and covariates were set as predictors and SMAR input parameters as predictands. Subsequently, we trained GAMs based on the site-optimized parameters and 161 347 different combinations of covariates. Finally, best combination models, with highest R^2 and lowest RMSE were chosen to predict SMAR parameters across space.

D. Study Area

This study was conducted in the southern Great Plains (SGPs) of the conterminous United States (CONUSs). SGP is one of the agricultural hubs of United States, producing grains such as wheat, sorghum, soybeans, and livestock. The SGP is on a relatively flat terrain with high climatic variability. The climate is characterized by aridity with mean annual rainfall ranging from 242 mm in the west to 1814 mm in the east. The mean daytime temperature ranges from approximately 14 °C in the winter to 40 °C in the summer; however, recent studies have shown a rise in temperature and aridity due to climate change [34]. The US Level III Ecoregion shows that the SGP covers prominent ecoregions including the High Plains, Central Great Plains, Cross Timbers, Flint Hills, Texas, and Blackland Prairies [35]. Recently, the region has experienced recurrent weather- and climate-related disasters, including droughts, floods, heat waves, and severe thunderstorms. A portion of the study region was subset for a detailed study. Fig. 1 shows the SGP and in situ observation stations spread across different land cover types.

E. In Situ Measurement

This study used in situ soil moisture measurements from the International Soil Moisture Network (ISMN) and Mesonet for the optimization and validation of the SMAR model, respectively. The ISMN is an open-source and harmonized global soil moisture database that provides on-site soil moisture at different depths and hydrometeorological data for validating satellite-based soil moisture products and calibrating land surface, climate, and hydrological models from 1952 to the present [36]. We chose 17 stations within our subset study area from three networks for optimizing the SMAR:

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ARM (n = 13), SCAN (n = 2), and USCRN (n = 2). The stations are mostly grassland and cultivated crops. Two depths were selected: 5 and 50 cm for SSM and RZSM, respectively. Hourly data were aggregated to the daily average. The individual measurements at depths between 5 and 50 cm were averaged to give measurements at 50 cm (2015–2019) because of variations of measurement depths; therefore, going deeper than 50 cm is not possible at some stations. The choice of 50 cm depth as rooting depth is justified as reported by White-Murillo et al. [2] and Liu et al. [37]. ISMN adopted automated quality control methods to ensure the quality of the datasets; and therefore, only measurements with ISMN quality flag "good" were used from 2015 to 2019. The database can be accessed at https://ismn.earth/en/. To ensure uniformity, the volumetric soil moisture was converted to relative soil moisture by dividing by the corresponding porosity (derived from PTFs). Similarly, six stations (grassland: 2; cultivated crop: 4) from the Oklahoma mesoscale network (Mesonet) were selected for validation. Mesonet is an automated network of 116 observation stations across Oklahoma, providing in situ soil moisture measurements every 30 min from 1991 to the present. The raw data were collected locally at each site and transferred remotely to the central ingest system at the Oklahoma Climatological Survey (OCS), University of Oklahoma, for further processing and quality checks [38]. Good-quality daily data (2017–2019) was obtained from the OCS and processed as the ISMN data. Additionally, meteorological data, including monthly precipitation and temperature were obtained from the Mesonet for calculating the SPEI.

F. Satellite-Based SSM

The current satellite-based soil moisture retrievals are produced at coarse spatial resolution (1–25 km) but with a fine temporal resolution of 1–2 days. However, retrievals at high spatiotemporal resolution are required to better understanding physical processes and applications in numerous fields. To overcome this challenge, the SMAP-HydroBlocks (henceforth, SMAP-HB) product representing soil moisture at surface layer (0–5 cm) was developed for the CONUS. The SMAP-HB product was generated using a merging framework, that can combine SMAP satellite microwave observations, in situ measurements, high-resolution land surface modeling, radiative transfer modeling, and machine learning to output a high-resolution soil moisture product (2015–2019). Evaluation of this product using 1191 in situ measurements showed a good correlation coefficient (R >0.8) and mean Kling–Gupta Efficiency (KGE = 0.56) at both field and watershed scales. Similarly, the SMAP-HB product outperformed the SMAP L3 and L4 products in capturing spatiotemporal variability at both the field and watershed scales [39]. We used the SMAP-HB SSM as an input to the SMAR model for predicting RZSM. The SMAP-HB product was downloaded from Zenodo (https://doi.org/10.5281/zenodo.5206725).

The US gridded soil properties and class 100 m were used in this study. This dataset was produced using machine learning algorithms, which combine point datasets from the National Cooperative Soil Survey (NCSS) characterization database, the National Soil Information System (NASIS), and the Rapid Carbon Assessment (RaCA) datasets, gSSURGO polygon maps, and over 200 soil covariates related to soil formation. The dataset is available at seven (7) standard soil depths (0, 5, 15, 30, 60, 100, and 200 cm). The dataset achieved 93% prediction accuracy for soil properties [40].

G. Soil Property Data

The percentage sand, clay, and bulk density maps were downloaded from (https://doi.org/10.18113/S1KW2H) at five soil depths (0, 5, 15, 30, and 60 cm). The percentage of silt was deduced from percent sand and clay, and field capacity and porosity were generated using Pedotransfer functions (PTFs) produced by Rawls et al. [32]. For the porosity, the average particle density was assumed to be 2.65 g/cm^3 [41]. For each of the soil property, 0–5 cm and 5–60 cm were averaged to obtain surface and root-zone soil properties, respectively.

Depth-to-bedrock map at 100-m resolution was obtained from the Soil Landscapes of the United States (SOLUS) repository at https://storage.googleapis.com/solus100pub/index.html. This product was developed using machine learning methods with 145 predictive soil-forming factor covariates. Cross-validation for the product yielded an R^2 of approximately 0.6 [42].

H. Evapotranspiration and Elevation Data

The AET data used in this study is the gap-filled MOD16A3GF v061 product (https://appeears.earthdatacloud.nasa.gov/). The product is generated annually at 500 m spatial resolution using Penman–Monteith equation and daily meteorological reanalysis data, and QC-checked MODIS data, including vegetation property dynamics, albedo, and land cover as inputs. Mu et al. [43] reported accuracy metrics for MODIS Global Evapotranspiration product: an average RMSE of 0.84 mm/day and a mean absolute error of 0.33 mm/day. The AET was used to derive water loss coefficient and as model's input parameters covariate.

NASA Shuttle Radar Topography Mission Global (SRTMGL3v003) elevation data at ~90 m spatial resolution was used in this study. The elevation data were used as also used as model's input parameters covariate. The AET and elevation data were accessed through the AppEEARS, application for extracting and exploring analysis-ready samples (https://appeears.earthdatacloud.nasa.gov/). The AET and elevation data were resampled using the bilinear interpolation to match the spatial resolution of the soil property data.

I. Land Cover Data

The National Land Cover Database (NLCD) produced by the Multi Resolution Land Characteristics Consortium (MRLC) was used to identify landcover types at the in situ measurement stations. The NLCD annual land cover product is produced from a long-term timeseries of Landsat Analysis Ready Data (ARD) using ensemble deep learning algorithms and postclassification accuracy. The product consists 16 classes of predominate land cover during the mapping year. The product was accessed from https://www.mrlc.gov/data

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TABLE I
USDM DROUGHT CLASSIFICATION

Class	Description	Percentile Range	SSMI
None	Normal or wet conditions	30.01 or Above	-0.49 or above
D0	Abnormally Dry	20.01 to 30.00	-0.5 to -0.79
D1	Moderate Drought	10.01 to 20.00	-0.8 to -1.29
D2	Severe Drought	5.01 to 10.00	-1.3 to -1.59
D3	Extreme Drought	2.01 to 5.00	-1.6 to -1.99
D4	Exceptional Drought	0.00 to 2.00	-2.0 or less

J. USDM Data and SSMI

The USDM is a project that releases weekly drought occurrence and severity maps across the CONUS since 1999. The methodology used to produce USDM maps is unique and popularly known as "a convergence of evidence approach," which blends multiscale physical data, drought impacts, and expert feedback and interpretations. The USDM classifications are shown in Table I: normal conditions, abnormally dry (D0), moderate (D1), severe (D2), extreme (D3), and exceptional (D4) (https://droughtmonitor.unl.edu/).

We collected the records of drought episodes occurrence from the USDM from 2016 to 2019 at the counties where our validation stations are located for comparison with the periods of suspected drought according to the SSMI (Table SII).

The SSMI is a dimensionless index used for drought monitoring. The monthly SSMI was calculated using the empirical Gringorten method. In this nonparametric method, the probability of the soil moisture data is determined empirically as $(i - 0.44)/(n + 0.12)$, where n and i refer to the number of observations and the rank of soil moisture dataset, respectively [44], [45]. Subsequently, the empirical probabilities were standardized to a normal distribution, and the SSMI was calculated as follows:

$$SSMI_{i,j} = \frac{SM_{i,j} - \overline{SM_j}}{\sigma_j} \tag{8}$$

where i is the data period range (2017 to 2019), j is month range (January to December); $\overline{SM_j}$ and σ_j are the mean and standard deviation of soil moisture in month j, respectively.

We calculated SSMI based on the predicted RZSM (SSMI-RZSM) and the SMAP-HB SSM (SSMI-SMAP-HB) at six validation stations to compare how well they capture drought events with respect to USDM observations.

Fig. 2 summarizes the methodological steps of the present study. Step 1 involves parameter optimization at the site, step 2 involves the prediction of the gridded parameters and the RZSM across space, and step 3 includes both assessments of the accuracy of the predicted RZSM and drought monitoring using RZSM.

K. Data Quality Check and Sensitivity Analysis

Spatiotemporal forecasting is susceptible to errors, arising from data and model uncertainties. Therefore, quantifying these errors improves the reliability and interpretability of predictions. Although we have shown, in the previous sections, the evaluation metrics of most of the data used, we rechecked the data quality where reliable ground-truth measurements were available. Systematic and random errors were quantified using validation and ubRMSE methods, respectively [46].

Similarly, to assess the sensitivity of the predicted RZSM to the uncertainties in the parameters, we ran a sensitivity analysis that tracks the difference in ubRMSE and RZSM outputs between the calibrated SMAR model and the SMAR models fit to perturbed inputs, since relating differences with respect to the performance and outputs of the calibrated SMAR model provides a consistent basis for comparison. Covariates were perturbed one at a time for each GAM-based SMAR parameter using a bootstrap with replacement approach and 1000 iterations. For each perturbation, we re-fit the GAM model using the previously chosen GAM structure for each SMAR parameter and then assessed the resulting ubRMSE and changed RZSM outputs based on running SMAR with the perturbed GAM covariate. The RZSM difference and ubRMSE difference for each iteration were calculated as follows:

$$RZSM_{diff,i} = RZSM_{orig,i} - RZSM_{adj,i} \tag{9}$$

$$ubRMSE_{diff,i} = ubRMSE_{orig,i} - ubRMSE_{adj,i} \tag{10}$$

where $RZSM_{orig,i}$ and $RZSM_{adj,i}$ denote the original and adjusted RZSM at iteration i, respectively. $ubRMSE_{orig,i}$ and $ubRMSE_{adj,i}$ denote the original and adjusted ubRMSE at iteration i.

III. RESULTS

A. Input Data Accuracy

The quality of the input data is paramount for the accuracy of RZSM predictions from SMAR. The SMAP-HB SSM correlated well with in situ measurements (r = ~ 0.70 to 0.89) at 88% of the sites, with ubRMSE of ≤ 0.06 cm^3/cm^3 at 94% of the sites (Table II). Generally, the soil property data (sand, silt, and clay) showed reasonable quality at both the surface and root-zone, with correlation coefficients ranging from ~0.40 to 0.70 and ~0.62 to 0.76, respectively. Similarly, the ubRMSE ranged from ~2.4% to 16.2% at the surface and ~5.4%–14.1% at the root-zone. On the other hand, elevation exhibited a very high correlation (r = ~ 0.95) with ubRMSE of ~15.3 m (Table SI). These results show that the essential SMAR input data can produce RZSM with reasonable accuracy.

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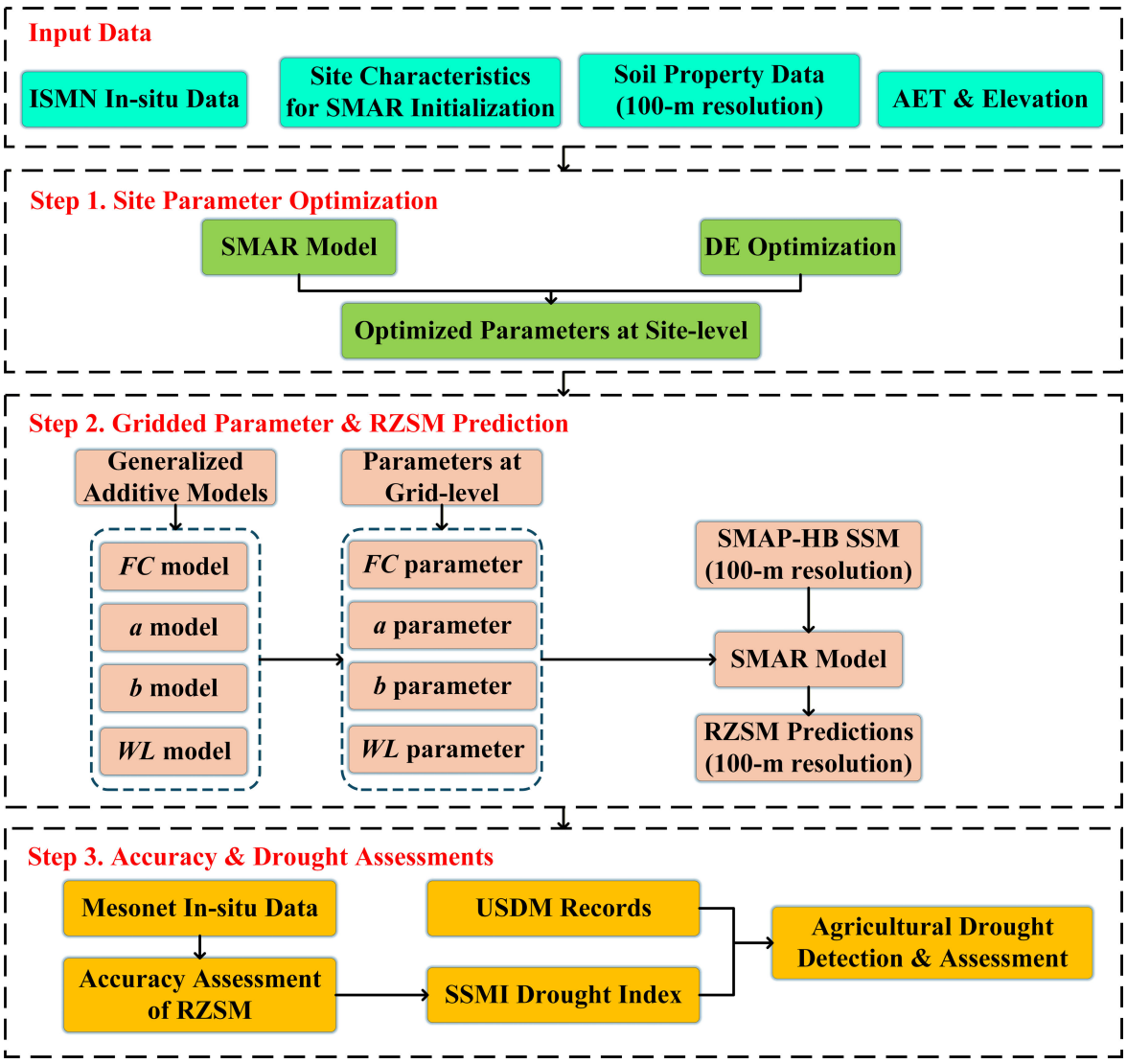


Fig. 2. Methodological flowchart of the study.

B. Optimized SMAR Model

SMAR model parameters are fundamental for inferring RZSM; therefore, their accuracy greatly influences the predicted RZSM accuracy. We optimized the SMAR parameters based on the accuracy between of the in situ observations and the predicted RZSM at sites. Fig. 3 shows the timeseries of relative in situ observations and predicted RZSM during model optimization. Fig. 3 shows that both the observed and predicted RZSM at the sites exhibit similar fluctuations over time, with peaks and valleys that align well. However, some deviations occurred at certain points, where the predicted values slightly differed from the observed values. Generally, the accuracy of the site parameters ranged from RMSE of 0.019–0.072 cm^3/cm^3, with an average of 0.039 cm^3/cm^3. This indicates that the predicted parameters at the sites have reasonable accuracy and met the NASA's quality benchmark of 0.06 cm^3/cm^3 (Table III).

To predict RZSM across space using SMAR, the parameters must be estimated at every pixel. Table IV shows the best models developed using GAM for predicting the spatial SMAR parameters. Where β_0 represents a global intercept, $f_j(x) = \sum_{m=1}^{kj} \beta_{jm} b_{jm}(x)$ denotes a smooth function, where k_j is the number of dimensions a smooth function has, which is related to degrees of freedom (df) for term, $j, b_{jm}(x)$ represents the basic smooth function (thin-plate regression spline), β_{jm} is a coefficient to be estimated, and ε_i is a random error term. Notably, the df controls flexibility and ensures computational stability in the models, thereby reducing the risk of overfitting. A df value = 1, indicates linear effect of a predictor, whereas a value of >1 indicates a nonlinear effect. Model FC_i maintains high nonlinear effects for all predictors (df = 3), except for elevation and BD_{surf_i}. Model α_i ensures moderate flexibility and focuses on the nonlinear effects of the predictors (df = 2–3). Conversely, models b_i and WL_i maintain explicitly both simple and nonlinear effects of the predictors (df = 1–3). These results demonstrate that all models reasonably capture both linear and nonlinear relationships between the predictors and response variables.

We tested many different structures and chose the GAM structure with the lowest RMSE for each SMAR parameter. Fig. S1 indicates how many GAM model structures we tested

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TABLE II
SMAP-HB SSM QUALITY CHECK AGAINST IN SITU OBSERVATIONS

Site	RMSE	ubRMSE	Bias	Error_SD	Correlation
Abrams	0.060	0.060	-0.003	0.060	0.361
Anthony	0.081	0.060	-0.055	0.060	0.690
Ashton	0.044	0.043	-0.008	0.043	0.828
Fort_Reno	0.078	0.051	-0.059	0.051	0.822
Lamont_CF	0.051	0.049	-0.015	0.049	0.815
Maple_City	0.107	0.073	-0.079	0.073	0.662
Marshall	0.046	0.044	-0.013	0.044	0.888
Medford	0.077	0.056	-0.053	0.056	0.782
Morrison	0.076	0.047	-0.059	0.047	0.865
Newkirk	0.102	0.086	0.055	0.086	0.683
Omega	0.045	0.043	-0.012	0.043	0.844
Pawhuska	0.078	0.045	-0.064	0.045	0.697
Ringwood	0.062	0.041	0.047	0.041	0.699
Stillwater_2_W	0.069	0.057	-0.039	0.057	0.804
Stillwater_5_WNW	0.046	0.036	0.029	0.036	0.871
Tryon	0.129	0.051	-0.118	0.051	0.816
Waukomis	0.052	0.050	-0.017	0.050	0.784

TABLE III
SITE SMAR PARAMETERS AND PREDICTION ACCURACY

Site	FC_{site}	a_{site}	b_{site}	WL_{site}	$RMSE_{site}$ (cm^3/cm^3)
Abrams	0.246	0.270	0.486	0.276	0.034
Anthony	0.121	0.549	0.315	0.281	0.019
Ashton	0.243	0.214	0.214	0.275	0.038
Fort_Reno	0.085	0.091	0.069	0.321	0.021
Lamont_CF	0.017	0.550	0.312	0.376	0.025
Maple_City	0.001	0.038	0.052	0.400	0.046
Marshall	0.145	0.030	0.058	0.377	0.048
Medford	0.224	0.033	0.067	0.400	0.046
Morrison	0.012	0.322	0.204	0.385	0.062
Newkirk	0.001	0.010	0.017	0.400	0.069
Omega	0.143	0.163	0.141	0.395	0.034
Pawhuska	0.243	0.083	0.122	0.285	0.032
Ringwood	0.038	0.489	0.435	0.039	0.025
Stillwater_2_W	0.020	0.448	0.328	0.368	0.033
Stillwater_5_WNW	0.070	0.536	0.454	0.166	0.024
Tryon	0.001	0.073	0.083	0.400	0.030
Waukomis	0.094	0.032	0.036	0.115	0.072
Average					**0.039**

and the RMSE at predicting each SMAR parameter. Each model shows a high level of accuracy in predicting pixel parameters, with an R^2 and RMSE ranging from 0.49 to 0.96 and 0.02–0.1, respectively (Table IV). To further evaluate the predictive models, the predicted RZSM using GAM-based predicted parameters (extracted pixel values at the sites) was tested against in situ observations. The accuracy of the gridded parameters ranged from RMSE of 0.031–0.268 cm^3/cm^3 across the sites with an average ubRMSE of 0.048 cm^3/cm^3, indicating that the GAM-based predictive models have reasonable accuracy (Table V). Finally, individual SMAR parameter maps were produced using GAM-based predictive models

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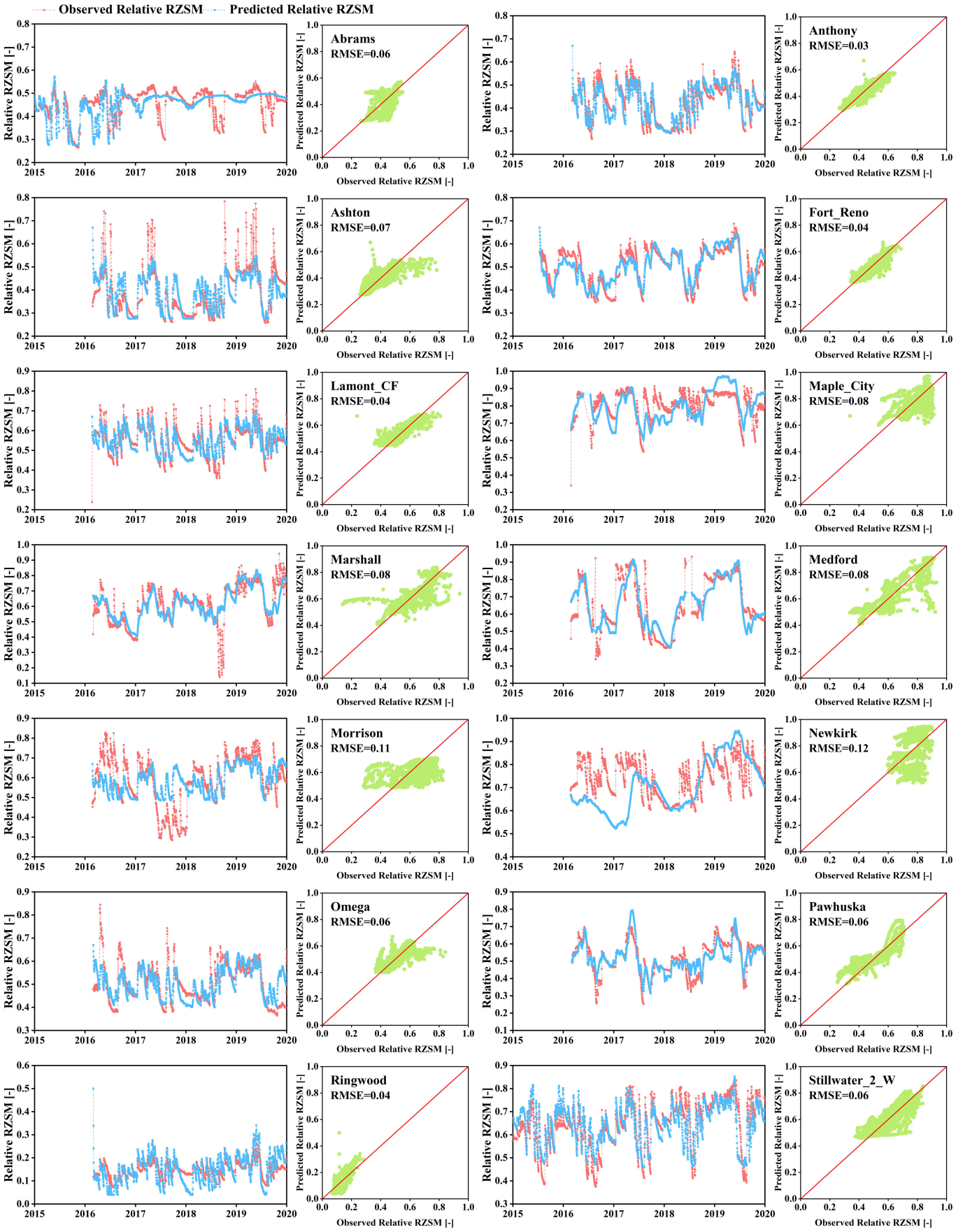


Fig. 3. Timeseries plot of predicted versus observed RZSM during model optimization.

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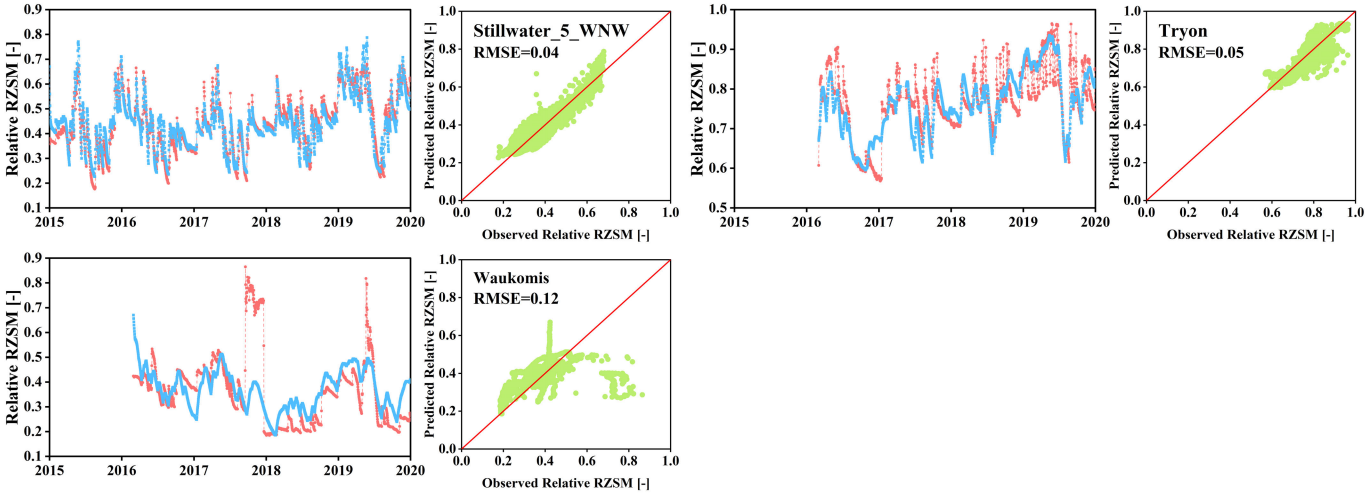


Fig. 3. *(Continued.)*Timeseries plot of predicted versus observed RZSM during model optimization.

TABLE IV
GAM FOR PREDICTING THE SMAR GRIDDED PARAMETERS

Model	R^2	RMSE
$FC_i = \beta_0 + \underbrace{\sum_{m=1}^{k_1=3} \beta_{1m} b_{1m}(BD_{root_i})}_{f_1} + \underbrace{\sum_{m=1}^{k_2=1} \beta_{2m} b_{2m}(BD_{surf_i})}_{f_2} + \underbrace{\sum_{m=1}^{k_3=3} \beta_{3m} b_{3m}(Depthtorock_i)}_{f_3} + \underbrace{\sum_{m=1}^{k_4=1} \beta_{3m} b_{3m}(Elevation_i)}_{f_4} + \underbrace{\sum_{m=1}^{k_5=3} \beta_{4m} b_{4m}(Sand_{surf_i})}_{f_5} + \underbrace{\sum_{m=1}^{k_6=3} \beta_{5m} b_{5m}(Sand_{root_i})}_{f_6} + \varepsilon_i$	0.86	0.03
$a_i = \beta_0 + \underbrace{\sum_{m=1}^{k_1=3} \beta_{1m} b_{1m}(AET_i)}_{f_1} + \underbrace{\sum_{m=1}^{k_2=2} \beta_{2m} b_{2m}(Depthtorocck_i)}_{f_2} + \underbrace{\sum_{m=1}^{k_3=3} \beta_{3m} b_{3m}(Elevation_i)}_{f_3} + \varepsilon_i$	0.81	0.10
$b_i = \beta_0 + \underbrace{\sum_{m=1}^{k_1=1} \beta_{1m} b_{1m}(AET_i)}_{f_1} + \underbrace{\sum_{m=1}^{k_2=1} \beta_{2m} b_{2m}(BD_{root_i})}_{f_2} + \underbrace{\sum_{m=1}^{k_3=3} \beta_{3m} b_{3m}(Elevation_i)}_{f_3} + + \underbrace{\sum_{m=1}^{k_4=3} \beta_{5m} b_{5m}(Sand_{root_i})}_{f_4} + \varepsilon_i$	0.85	0.06
$WL_i = \beta_0 + \underbrace{\sum_{m=1}^{k_1=1} \beta_{2m} b_{2m}(BD_{surf_i})}_{f_1} + \underbrace{\sum_{m=1}^{k_2=3} \beta_{3m} b_{3m}(Clay_{surf_i})}_{f_2} + \varepsilon_i$	0.49	0.07

were shown in Fig. 4. It is evident that the parameters varied distinctively across the space, indicating the capability of the models to capture spatial variability.

Results of the correlation between the site and gridded parameters showed a strong relationship, with an R^2 ranging from 0.48 to 0.95, and an RMSE ranging from 0.02 to 0.12 (Fig. 5). The relative field capacity was less sensitive to the scale difference with smallest RMSE of 0.02. In contrast, water loss coefficient showed significant bias with an RMSE of 0.12, indicating that the water loss coefficient was very sensitive to scale difference. Therefore, the diffusion coefficient, relative field capacity, and relative wilting level were less sensitive to the scale difference compared with water loss coefficient. On the whole, 65% of the sites showed good agreement between the site and gridded parameters, with an RMSE difference of 0.06 and below, indicating an acceptable accuracy.

The gridded parameters were subsequently used to produce RZSM maps at 100 m resolution. The RZSM varied seasonally, with the highest values occurring in spring and winter while summer and autumn showed more dry zones, especially in the southern and central regions (Fig. S2). As reported by Flanagan [47], the SGP received the highest precipitation in the mid-to-late spring and less in summer months, thereby increasing the chances of temperature rise and consequently, the occurrence of drought episodes in summer. Therefore, this result indicates that GAM-based SMAR can adequately capture RZSM's seasonal variability.

C. Accuracy Assessment of the Predicted RZSM

To assess the accuracy of the GAM-based spatial SMAR, the predicted RZSM was validated against in situ observations at six Mesonet stations. Fig. 6 shows six scatter plots with evaluation metrics for each station. The R^2 ranged from 0.479 at the BREC station to 0.716 at the LAHO station, indicating a strong and moderate correlation between the in situ observation and predicted measurements across the stations. All stations produced RMSE values below 0.06 cm^3/cm^3, except BREC (RMSE = 0.067 cm^3/cm^3), which is the NASA's recommended quality benchmark for predicting soil moisture with satellite observations. The ubRMSE ranged from 0.016 to 0.040 cm^3/cm^3, indicating small systematic prediction error at all stations. Overall, the spatial SMAR accurately predicted RZSM values across all stations, with the highest accuracy in LAHO and REDR, and moderate accuracy in BLAC, CARL, BREC, and ELR.

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TABLE V
SMAR GRIDDED-PARAMETERS AND PREDICTION ACCURACY

Site	FC_{grid}	a_{grid}	b_{grid}	WL_{grid}	RMSE$_{grid}$ (cm^3/cm^3)	ubRMSE$_{grid}$ (cm^3/cm^3)
Abrams	0.259	0.133	0.435	0.297	0.268	0.063
Anthony	0.138	0.517	0.353	0.252	0.049	0.040
Ashton	0.199	0.043	0.079	0.337	0.249	0.046
Fort_Reno	0.036	0.091	0.056	0.302	0.031	0.030
Lamont_CF	0.005	0.190	0.165	0.288	0.085	0.039
Maple_City	0.003	0.066	0.091	0.314	0.059	0.059
Marshall	0.152	0.038	0.056	0.309	0.069	0.045
Medford	0.196	0.083	0.122	0.300	0.057	0.057
Morrison	0.007	0.216	0.099	0.298	0.089	0.069
Newkirk	0.004	0.035	0.025	0.343	0.074	0.037
Omega	0.117	0.073	0.090	0.344	0.095	0.041
Pawhuska	0.222	0.071	0.132	0.302	0.167	0.022
Ringwood	0.037	0.377	0.398	0.043	0.158	0.064
Stillwater_2_W	0.039	0.517	0.437	0.342	0.079	0.038
Stillwater_5_WNW	0.046	0.355	0.307	0.303	0.176	0.034
Tryon	0.001	0.114	0.194	0.336	0.097	0.043
Waukomis	0.103	0.103	0.067	0.294	0.102	0.081
Average					**0.112**	**0.048**

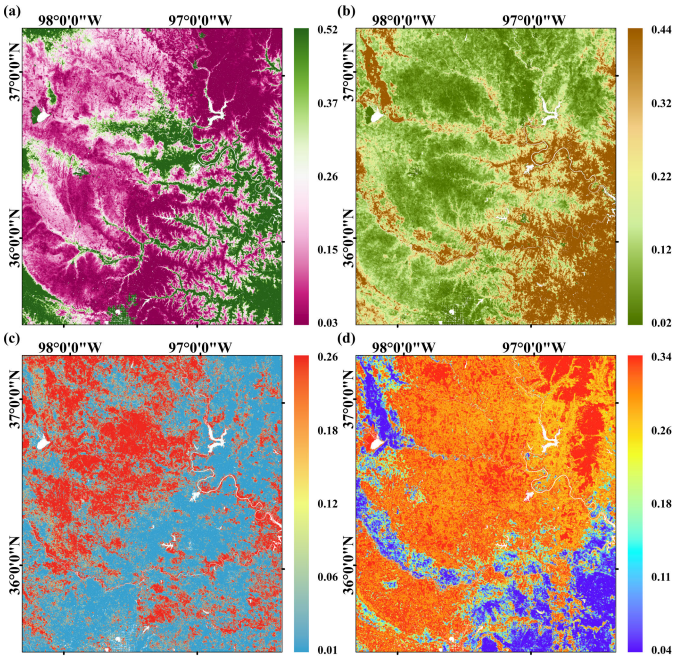


Fig. 4. Spatial distribution of the SMAR parameters at a resolution of 100 m. (a) Water loss coefficient, (b) diffusion coefficient, (c) relative field capacity, and (d) relative wilting level.

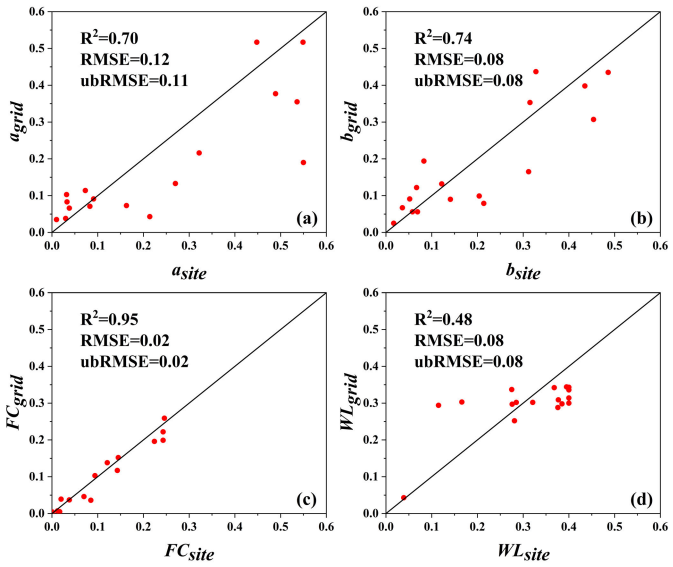


Fig. 5. Scatterplots between the site and gridded SMAR parameters. (a) Water loss coefficient, (b) diffusion coefficient, (c) relative field capacity, and (d) relative wilting level.

A scatterplot of combined data from all stations was produced to assess the cumulative accuracy of the spatial SMAR across all stations (Fig. 7). The evaluation metrics indicate that the spatial GAM-based SMAR has good predictive capabilities with an overall RMSE of 0.043 cm^3/cm^3 and bias of 0.017 cm^3/cm^3. However, the combined R^2 was reduced to 0.501, due to the presence of few outliers, accounting for increased noise and station heterogeneity. Consequently, the R^2 became more sensitive to outliers when data from all stations were combined.

D. Predicted RZSM for Monitoring Agricultural Drought

Soil moisture particularly at the rooting depth, is essential for drought monitoring. We compared the monthly SSMI-RZSM and SSMI-SMAP-HB with respect to the USDM observations to assess the performance of the predicted RZSM in capturing drought episodes. Fig. 8 shows the evolution of SSMI over the study period with −0.8, marked as a reference line for drought episode (Table I). Both the SSMI-RZSM and SSMI-SMAP-HB demonstrated a reasonable relationship with respect to records of drought episode occurrence from the USDM, 2017–2019 (Table SII) at the counties where the validation sites are located. According to the findings, the

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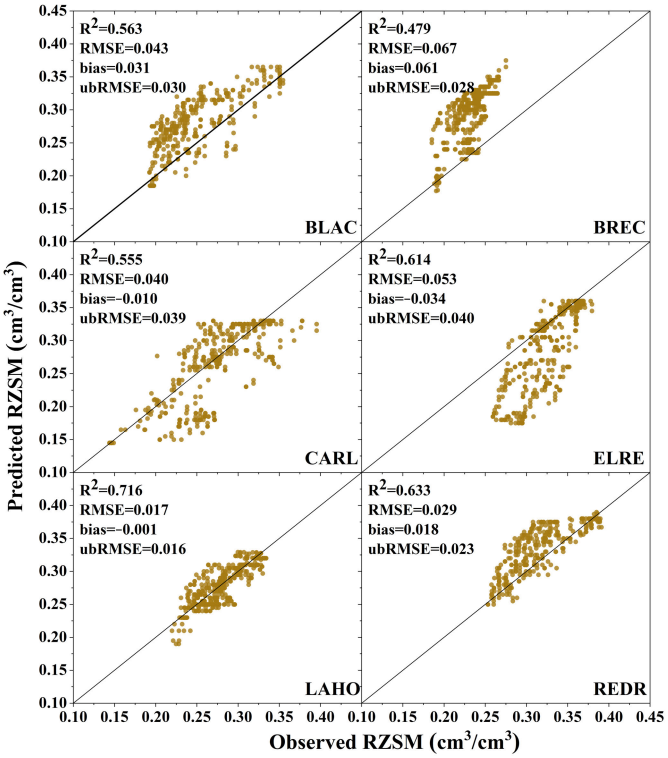


Fig. 6. Accuracy of the predicted RZSM at individual Mesonet stations.

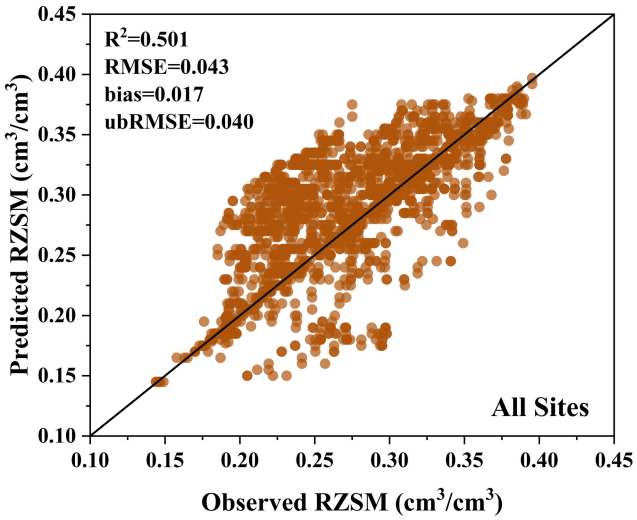


Fig. 7. Accuracy of the predicted RZSM at all combined Mesonet stations.

SSMI-RZSM captured 53%–90% of the drought episodes, with an average of 71% across the validation stations from 2017 to 2019. Moreover, a comparison of the SSMI-RZSM and SSMI-SMAP-HB in terms of catching drought episodes revealed that the former was better than the latter, which only captured 33%–64% of drought episodes, with an average of 46% (Fig. 8). Across all validation sites, the SSMI-RZSM outperformed SSMI-SMAP-HB in terms of capturing drought events, with a notable difference ranging from 13% to 33%.

IV. Discussion

A. Model Accuracy

The goal of the predictive models is to provide estimates with reasonable accuracy. In this study, the GAM-based SMAR model produced reasonably accurate RZSM estimates at the site level and across space during the optimization and validation periods. The overall RMSE values for the optimization and validation periods were 0.039 and 0.043 cm^3/cm^3, respectively. This result shows that the GAM-based SMAR predictions of RZSM met the NASA's quality benchmark for predicting soil moisture from satellite observations (RMSE = 0.06 cm^3/cm^3). It is noteworthy that the accuracy of SMAR model relies to some extent on the initial conditions. We noted that the high relative root-zone field capacity resulted in RZSM overestimation, which ultimately decreased the prediction accuracy. Similarly, high root-zone clay content can result in overestimation as RZSM predictions reached as high as 0.6 cm^3/cm^3 (Fig. S2) in areas with high root-zone clay content (Fig. S3). This corroborated the findings of Baldwin et al. [6] who reported that the initial conditions for the SMAR model can greatly affect its accuracy. Additionally, a good representation of soil property is of paramount importance for the accuracy of RZSM predictions from SMAR.

The results of the sensitivity analysis showed that the predicted RZSM was sensitive to the uncertainties in the SMAR input parameters. We found that perturbing root-zone sand content, elevation, and depth to bedrock for the "*a*" and "*b*" SMAR parameters had the greatest impact on ubRMSE and RZSM outputs (Figs. S4–5). Overall, ubRMSE differences ranged from about −0.032 to 0.028 cm^3/cm^3 at the extremes, and at least 95% of absolute ubRMSE differences were <0.01 cm^3/cm^3. RZSM outputs are most sensitive to depth to bedrock perturbations for the "*a*" SMAR parameter and root-zone sand content perturbations for the "*b*" SMAR parameter. The 95th percentile of all absolute RZSM differences across the simulations is 0.087 cm^3/cm^3. In addition, we note that the observed standard deviation of daily volumetric RZSM content is 0.085 cm^3/cm^3, which is higher than the standard deviation of the differences between volumetric RZSM predicted by the SMAR model with parameters predicted by our chosen GAM models and the RZSM predicted during all iterations of the sensitivity analysis for each input parameter perturbed (max) SD of RZSM differences = 0.063 cm^3/cm^3 from perturbing elevation in the "*a*" parameter GAM model.

Despite all sources of SMAR model uncertainties, our approach produced a higher resolution RZSM 100-m spatial resolution at daily timestep and appeared to be more accurate than some previous studies (e.g., [22], [48]); however, they attributed the low accuracy to poor representation of regional parameters and coarse spatial resolution of the SSM (1–25 km), which is susceptible to noises.

B. Model Assumption

It is worth noting to emphasize that SMAR model operates on the assumption that the water flux from the surface layer is inconsequential except that it exceeds field capacity and percolation occurs instantaneously with a linear water loss function. In practice, it has been found that soil water loss

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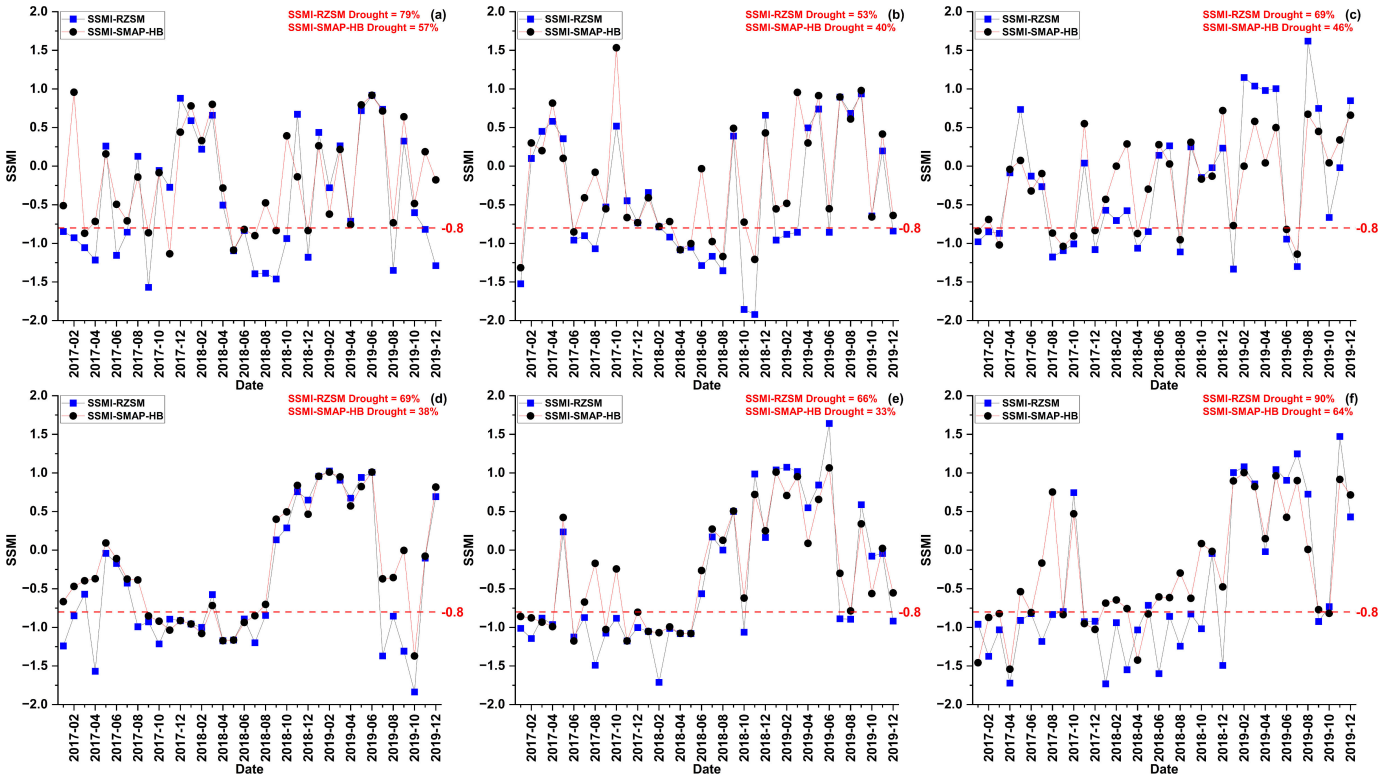


Fig. 8. Monthly SSMI-RZSM versus SSMI-SMAP-HB, with percentage of drought occurrence captured (2017–2019) based on USDM records. (a) BLAC, (b) BREC, (c) CARL, (d) ELRE, (e) LAHO, and (f) REDR. Note that both the SSMI-RZSM and SSMI-SMAP-HB are at 100 m resolution.

function is highly nonlinear in humid regions where soils are well watered and is more pronounced in finer soil texture with higher water retention capacity and lower infiltration rate [49]. Therefore, the SMAR performed better in arid and semiarid regions with flat surfaces. The chosen study area is suitable for application of the SMAR because it is water-limited and relatively flat area with soil texture ranging from loam to sandyloam. However, the poor performance of the model during optimization at Morrison, Newkirk, and Waukomis sites can be attributed to the model's sources of uncertainty, such as unaccounted lateral flow and capillary rise in its structure. Similarity, it is important to mention that the SMAR model's assumptions were built upon the exponential filter approach, which is the foundation for SMOS-L4 RZSM product at 25 km resolution. Therefore, the applicability of the SMAR model at resolutions ranging from 100 m to 25 km is valid despite the heterogeneity of the surface conditions, including soil properties and landcover.

C. Relationship Between SMAR Parameters and Covariates

The SMAR model parameters are a solid foundation for predicting RZSM. For the first time, we demonstrated the predictive capability of GAM models in predicting spatial SMAR parameters at 100-m spatial resolution. Consequently, the RZSM was predicted at an unprecedented spatial resolution of 100 m using the SMAR model. This is a significant development toward predicting RZSM at resolutions higher than 100 m, as long as soil property data and SSM are available. Remarkably, GAM produced best-fitting models that are in conformity with findings of previous studies and soil physics. The relative field capacity is strongly related to both the surface and root-zone bulk density, indicating that porosity determines how soon the soil can attain the saturation level. Similarly, both surface and root-zone sand fractions were found to be related to the relative field capacity, as coarse-textured soils readily attain saturation. The close link between the relative field capacity and sand fraction is obvious, as it has been reported that it increases with small pore spaces and decreases with large pore spaces [50]. Water loss was strongly related with the depth to bedrock and AET. This result agreed with the findings of previous studies [21]. It is quite interesting that the water loss is strongly related to AET as the soil water is lost through evaporation from bare soils and transpiration from plants, and this is one of the basic assumptions of the SMAR formulation [20]. The diffusion coefficient was strongly related to AET, root-zone bulk density, and sand fraction. This is an important finding because a high diffusion coefficient indicates high moisture content at the root-zone, which in turn implies less water loss through evapotranspiration and high porosity, which is related to bulk density and coarse-textured soils (e.g., sands). This result is corroborated by previous studies [21]. The relative wilting level was strongly related to surface bulk density and clay fraction. Moreover, previous studies have shown that the clay fraction and bulk density play significant role in determining the water-holding capacity of soils [51].

It is noteworthy that elevation was strongly related to all SMAR parameters except the relative wilting level. Baldwin et al. [6] found that the accuracy of the spatial

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TABLE VI
CALENDAR OF GROWING SEASON AND CROP WINDOW FOR THE STUDY AREA

Crop	Planting period	Harvest period	Crop window	Growth stage
Winter wheat	Sept. - Nov.	Jun. – Jul.	May - Jun.	Grain filling and kernel development
Corn	Mar. -May.	Aug. – Oct.	Jun. – Aug	Tasseling and grain fill
Soybean	Apr.-Jul.	Oct. – Nov.	Jun. – Sept.	Flowering to pod fill
Sorghum	Apr.-Jul.	Sept. – Nov.	Apr. – Jul.	Early growth
Cotton	May-Jun.	Oct. – Dec.	Jun. – Aug.	Flowering and boll development

SMAR parameters was drastically reduced when elevation was removed from the predictors using a neural network, indicating that it plays a significant role in soil water dynamics. One critical issue in estimating SMAR parameters that needs to be elaborated is that the relative field capacity is much smaller at some sites than the relative wilting level (e.g., [20], [21]), which may be confusing from the soil physics point of view. The relative field capacity and wilting level differ technically from the absolute measurements. The former is associated with the near-surface layer (0–2 or 5 cm), whereas the latter is associated with the entire root-zone. Consequently, the relative wilting level tends to be greater than the relative field capacity. The most important thing is the final output of the RZSM from the SMAR, which is reasonably accurate in the previous and present studies.

D. Drought Monitoring

Drought monitoring and early warning is the first and key step in integrated drought monitoring. Many studies have demonstrated a good agreement between in situ and modeled-derived soil moisture estimates at rooting depth and drought indices; hence, it represents agricultural drought [27], [52]. Additionally, soil moisture-based drought indices outperformed other indices derived from meteorological variables [53]. Similarly, the results of this study showed a positive relationship between a multivariate drought index, the USDM, and the SSMI-RZSM at a depth of 50 cm. Moreover, comparison between the SSMI-RZSM and SSMI-SMAP-HB in depicting drought episodes showed that the former outperformed the latter at six Mesonet sites. These results are in agreement with the findings of Hao and Aghakouchak [54], who reported temporal consistency of SSMI computed from satellite-based profile soil moisture and USDM observations. Xu et al. [25] also reported that SSMI, computed from assimilated based soil moisture at 10 cm consistently detects drought episodes in comparison with USDM Observations. Additionally, some studies reported that RZSM is a good representation of drought indices. Tian et al. [52] found that the in situ soil moisture at 0–60 cm depth reasonably correlated with six drought indices for monitoring agricultural drought, with SPEI-1 being the most representative of soil moisture in the southern US. On the other hand, Ford and Quiring [27] reported that model-derived soil moisture (NLDAS-2 VIC) at 10–40 cm depth captured drought events better than at 0–10 cm layers. The superiority of RZSM in capturing agricultural drought is not surprising because RZSM greatly controls evapotranspiration rate, which can indicate moisture stress at root-zone soil layers and subsequently provide useful information on agricultural drought [27].

Similarly, identification of drought during the sensitive crop growth stage (crop window) in the growing season calendar (Table VI) is of paramount importance to growers. An overlay of the crop calendar for five major crops grown in the study area and the SSMI-RZSM and SSMI-SMAP-HB revealed that the SSMI-RZSM captured moisture deficit better than SSM during the crop windows at six Mesonet sites (Fig. 8). For example, the most consistently drought-vulnerable crops were winter wheat, corn, and sorghum (Figs. S6–11) because of the consistent moisture deficit at the root-zone during planting and harvest periods. This result agreed well with the USDM, which reported that winter wheat was affected by droughts with noticeable impacts such as decoloration, slow growth, and ultimately reduced yields during growing seasons (2016–2019) (Table SIII). Conversely, soybean and cotton were likely to be less affected by drought impacts because of relatively stable moisture content during their sensitive growth stages, except in the REDR site where a noticeable moisture deficit was observed. These findings indicate that the SSMI-RZSM can represent agricultural drought.

Moreover, the ability of a drought predictor to accurately identify the beginning and end of drought episodes is critical in drought monitoring and early warning systems [55]. Interestingly, our results showed that the SSMI-RZSM adequately depicted the onset and cessation of drought events. According the USDM records, most of the counties in the study were under moderate to severe droughts between January and July 2018, which SSMI-RZSM reasonably captured (Table SIII). This is a useful result, as the SSMI-RZSM can be used instead of multivariate drought indices that are computationally demanding and data-intensive.

E. Limitations

The GAM-based SMAR model is promising in predicting RZSM from SSM and soil properties; however, the accuracy of the relative root-zone field capacity and estimated porosity are a critical concern for the application of SMAR. The latter converts the relative moisture to volumetric water content, and the former initiates the SMAR model. Currently, these measurements are either pulled from the experimental results on different soil textures as reported by Rawls et al. [32] or produced using PTFs. In most cases, the estimated porosity is too low to covert relative RZSM to reasonable volumetric

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measurements without bumping up, as in Baldwin et al. [6] and the present study. Most of the hydraulic properties of soil are estimated and generalized from the soil water retention curves. According to Twarakavi et al. [56], field capacity is intrinsically connected with dynamic natural processes; therefore, estimating it from retention curves based on benchmark pressure heads is highly liable to errors. Therefore, more accurate and reliable measurements of the root-zone field capacity and porosity are essential for accurate RZSM predictions using SMAR model.

V. Conclusion

Regional estimation of RZSM is essential as it directly or indirectly controls energy and water fluxes among soil, plant, and atmosphere, which is useful for smart agriculture, drought monitoring, and watershed management. In this study, we predicted RZSM at 100-m spatial resolution and daily timestep for the first time using the GAM-based SMAR infiltration model at a portion of the US SGPs. A differential evolution algorithm was used to optimize the SMAR parameters at sites, which are the foundation for running the model. Finally, the RZSM was predicted across space from 2015 to 2019. The results showed that the GAM-based spatial SMAR captured the seasonal variability of the RZSM during the study period. For the first time, we evaluated how accurate the SSMI, produced from the predicted RZSM can capture agricultural drought. The results showed that an average of 71% of the drought occurrences were captured by the SSMI-RZSM, while the SSMI-SMAP-HB only detected an average of 46% of drought events during the study period. In conclusion, we demonstrated the potential of the GAM-based spatial SMAR model for predicting RZSM across space at high spatiotemporal resolution, which is a crucial step toward estimating higher-resolution regional RZSM for regional agricultural drought monitoring and water management. Nevertheless, we also came to the conclusion that more work is required to obtain accurate measurements of the root-zone field capacity and porosity at the point scale and across space, as these measurements are essential in the application of proposed approach.

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References

[1] H. Gao, M. Hrachowitz, S. J. Schymanski, F. Fenicia, N. Sriwongsitanon, and H. H. G. Savenije, "Climate controls how ecosystems size the root zone storage capacity at catchment scale," *Geophys. Res. Lett.*, vol. 41, no. 22, pp. 7916–7923, Nov. 2014, doi: 10.1002/2014gl061668.

[2] L. F. White-Murillo et al., "Towards resolution enhancement of P-band brightness temperature data using passive-passive downscaling with L-band radiometer data," *Remote Sens. Environ.*, vol. 323, Jun. 2025, Art. no. 114737, doi: 10.1016/j.rse.2025.114737.

[3] N. Romano, C. Mazzitelli, and P. Nasta, "Root-zone water-storage capacity and uncertainty: An intrinsic factor affecting agroecosystem resilience to drought," *Water Resour. Res.*, vol. 61, no. 1, p. 2024, Jan. 2025, doi: 10.1029/2024wr037719.

[4] A. A. Kasim et al., "Remote sensing of root zone soil moisture: A review of methods and products," *J. Hydrol.*, vol. 656, Aug. 2025, Art. no. 133002, doi: 10.1016/j.jhydrol.2025.133002.

[5] J. G. B. Leenaars et al., "Mapping rootable depth and root zone plant-available water holding capacity of the soil of Sub-Saharan Africa," *Geoderma*, vol. 324, pp. 18–36, Aug. 2018, doi: 10.1016/j.geoderma.2018.02.046.

[6] D. Baldwin, S. Manfreda, H. Lin, and E. A. H. Smithwick, "Estimating root zone soil moisture across the eastern United States with passive microwave satellite data and a simple hydrologic model," *Remote Sens.*, vol. 11, no. 17, p. 2013, Aug. 2019, doi: 10.3390/rs11172013.

[7] R. Rivieccio, C. Di Bene, M. Paolanti, M. Marchetti, and R. Napoli, "Soil rooting depth of Italy," *J. Maps*, vol. 16, no. 2, pp. 36–42, Dec. 2020, doi: 10.1080/17445647.2019.1690595.

[8] C. Carranza, C. Nolet, M. Pezij, and M. van der Ploeg, "Root zone soil moisture estimation with random forest," *J. Hydrol.*, vol. 593, Feb. 2021, Art. no. 125840, doi: 10.1016/j.jhydrol.2020.125840.

[9] M. G. Grillakis, A. G. Koutroulis, D. D. Alexakis, C. Polykretis, and I. N. Daliakopoulos, "Regionalizing root-zone soil moisture estimates from ESA CCI soil water index using machine learning and information on soil, vegetation, and climate," *Water Resour. Res.*, vol. 57, no. 5, May 2021, Art. no. e2020WR029249, doi: 10.1029/2020wr029249.

[10] A. M. Peterson, W. D. Helgason, and A. M. Ireson, "Estimating field-scale root zone soil moisture using the cosmic-ray neutron probe," *Hydrol. Earth Syst. Sci.*, vol. 20, no. 4, pp. 1373–1385, Apr. 2016, doi: 10.5194/hess-20-1373-2016.

[11] T. N. Carlson, "The right triangle model: Overcoming the sparse data problem in Thermal/Optical remote sensing of soil moisture," *Remote Sens.*, vol. 16, no. 17, p. 3231, Aug. 2024, doi: 10.3390/rs16173231.

[12] K. Przeździecki, J. J. Zawadzki, M. Urbaniak, K. Ziemblińska, and Z. Miatkowski, "Using temporal variability of Land Surface Temperature and normalized vegetation index to estimate soil moisture condition on forest areas by means of remote sensing," *Ecol. Indicators*, vol. 148, Apr. 2023, Art. no. 110088, doi: 10.1016/j.ecolind.2023.110088.

[13] R. H. Reichle et al., "Version 4 of the SMAP level-4 soil moisture algorithm and data product," *J. Adv. Model. Earth Syst.*, vol. 11, no. 10, pp. 3106–3130, Oct. 2019, doi: 10.1029/2019ms001729.

[14] W. Wagner et al., "The ASCAT soil moisture product: A review of its specifications, validation results, and emerging applications," *Meteorologische Zeitschrift*, vol. 22, no. 1, pp. 5–33, Feb. 2013, doi: 10.1127/0941-2948/2013/0399.

[15] B. Martens et al., "GLEAM v3: Satellite-based land evaporation and root-zone soil moisture," *Geoscientific Model Develop.*, vol. 10, no. 5, pp. 1903–1925, May 2017, doi: 10.5194/gmd-10-1903-2017.

[16] O. Sungmin and R. Orth, "Global soil moisture data derived through machine learning trained with in-situ measurements," *Sci. Data*, vol. 8, no. 1, Jul. 2021, Art. no. 170, doi: 10.1038/s41597-021-00964-1.

[17] Z. Xing et al., "A first assessment of satellite and reanalysis estimates of surface and root-zone soil moisture over the permafrost region of Qinghai–Tibet Plateau," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112666, doi: 10.1016/j.rse.2021.112666.

[18] S. Yang, J. Zeng, W. Fan, and Y. Cui, "Evaluating root-zone soil moisture products from GLEAM, GLDAS, and ERA5 based on in situ observations and triple collocation method over the Tibetan Plateau," *J. Hydrometeorol.*, vol. 23, no. 12, pp. 1861–1878, Dec. 2022, doi: 10.1175/jhm-d-22-0016.1.

[19] L. Xu, N. Chen, X. Zhang, H. Moradkhani, C. Zhang, and C. Hu, "In-situ and triple-collocation based evaluations of eight global root zone soil moisture products," *Remote Sens. Environ.*, vol. 254, Mar. 2021, Art. no. 112248, doi: 10.1016/j.rse.2020.112248.

[20] S. Manfreda, L. Brocca, T. Moramarco, F. Melone, and J. Sheffield, "A physically based approach for the estimation of root-zone soil moisture from surface measurements," *Hydrol. Earth Syst. Sci.*, vol. 18, no. 3, pp. 1199–1212, Mar. 2014, doi: 10.5194/hess-18-1199-2014.

[21] D. Baldwin, S. Manfreda, K. Keller, and E. A. H. Smithwick, "Predicting root zone soil moisture with soil properties and satellite near-surface moisture data across the conterminous United States," *J. Hydrol.*, vol. 546, pp. 393–404, Mar. 2017, doi: 10.1016/j.jhydrol.2017.01.020.

[22] M. Farokhi, F. Faridani, R. Lasaponara, H. Ansari, and A. Farid-hosseini, "Enhanced estimation of root zone soil moisture at 1 km resolution using SMAR model and MODIS-based downscaled AMSR2 soil moisture data," *Sensors*, vol. 21, no. 15, p. 5211, Jul. 2021, doi: 10.3390/s21155211.

[23] S. Feng, S. Zhao, X. Huang, X. Guo, J. Fan, and Z. Zhang, "A novel standardized optimal synthesized drought index (SOSDI) for agro-drought monitoring using multisource remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5007516, doi: 10.1109/TGRS.2025.3608277.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:11:41 UTC from IEEE Xplore. Restrictions apply.

[24] Z. Hou and Z. Pu, "Assessing CYGNSS satellite soil moisture data for drought monitoring with multiple datasets and indicators," *Remote Sens.*, vol. 16, no. 1, p. 116, Dec. 2023, doi: 10.3390/rs16010116.
[25] L. Xu, P. Abbaszadeh, H. Moradkhani, N. Chen, and X. Zhang, "Continental drought monitoring using satellite soil moisture, data assimilation and an integrated drought index," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112028, doi: 10.1016/j.rse.2020.112028.
[26] S. Yuan, S. M. Quiring, and C. Zhao, "Evaluating the utility of drought indices as soil moisture proxies for drought monitoring and land-atmosphere interactions," *J. Hydrometeorol.*, vol. 21, no. 9, pp. 2157–2175, Sep. 2020, doi: 10.1175/jhm-d-20-0022.1.
[27] T. W. Ford and S. M. Quiring, "Comparison of contemporary in situ, model, and satellite remote sensing soil moisture with a focus on drought monitoring," *Water Resour. Res.*, vol. 55, no. 2, pp. 1565–1582, Feb. 2019, doi: 10.1029/2018wr024039.
[28] M. F. Ahmad, N. A. M. Isa, W. H. Lim, and K. M. Ang, "Differential evolution: A recent review based on state-of-the-art works," *Alexandria Eng. J.*, vol. 61, no. 5, pp. 3831–3872, May 2022, doi: 10.1016/j.aej.2021.09.013.
[29] R. Storn and K. Price, "Differential evolution—A simple and efficient heuristic for global optimization over continuous spaces," *J. Global Optim.*, vol. 11, no. 4, pp. 341–359, Dec. 1997.
[30] X. Wang and X. Yu, "Differential evolution algorithm with three mutation operators for global optimization," *Mathematics*, vol. 12, no. 15, p. 2311, Jul. 2024, doi: 10.3390/math12152311.
[31] K. Mullen, D. Ardia, D. Gil, D. Windover, and J. Cline, "DEoptim: AnRPackage for global optimization by differential evolution," *J. Stat. Softw.*, vol. 40, no. 6, pp. 1–26, 2011, doi: 10.18637/jss.v040.i06.
[32] W. J. Rawls, L. R. Ahuja, D. L. Brakensiek, and A. Shirmohammadi, "Infiltration and water movement," in *Handbook of Hydrology*. New York, NY, USA: McGraw-Hill, 1993.
[33] Z.-W. Zhao, P. Leng, X.-J. Han, and G.-F. Shang, "Global maize yield responses to essential climate variables: Assessment using atmospheric reanalysis and future climate scenarios," *Comput. Electron. Agricult.*, vol. 232, May 2025, Art. no. 110140, doi: 10.1016/j.compag.2025.110140.
[34] X. Yang, X. Xiao, Y. Qin, J. Wang, and K. Neal, "Mapping forest in the southern great plains with ALOS-2 PALSAR-2 and Landsat 7/8 data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 104, Dec. 2021, Art. no. 102578, doi: 10.1016/j.jag.2021.102578.
[35] J. M. Omernik and G. E. Griffith, "Ecoregions of the conterminous United States: Evolution of a hierarchical spatial framework," *Environ. Manage.*, vol. 54, no. 6, pp. 1249–1266, Dec. 2014, doi: 10.1007/s00267-014-0364-1.
[36] W. A. Dorigo et al., "The international soil moisture network: A data hosting facility for global in situ soil moisture measurements," *Hydrol. Earth Syst. Sci.*, vol. 15, no. 5, pp. 1675–1698, May 2011, doi: 10.5194/hess-15-1675-2011.
[37] Y. Liu, T. Li, W. Zhang, and A. Lv, "Regionalization of root zone moisture estimations from downscaled surface moisture and environmental data with the soil moisture analytical relationship model," *Water*, vol. 15, no. 23, p. 4133, Nov. 2023, doi: 10.3390/w15234133.
[38] B. G. Illston et al., "Mesoscale monitoring of soil moisture across a statewide network," *J. Atmos. Ocean. Technol.*, vol. 25, no. 2, pp. 167–182, Feb. 2008, doi: 10.1175/2007jtecha993.1.
[39] N. Vergopolan et al., "SMAP-HydroBlocks, a 30-m satellite-based soil moisture dataset for the conterminous U.S.," *Sci. Data*, vol. 8, no. 1, p. 264, Oct. 2021, doi: 10.1038/s41597-021-01050-2.
[40] A. Ramcharan et al., "Soil property and class maps of the conterminous United States at 100-meter spatial resolution," *Soil Sci. Soc. Amer. J.*, vol. 82, no. 1, pp. 186–201, Jan. 2018, doi: 10.2136/sssaj2017.04.0122.
[41] O. Boiko, S. Kagone, and G. Senay, "Soil properties dataset in the United States," *U.S. Geological Surv.*, 2021, doi: 10.5066/P9TI3IS8.
[42] T. W. Nauman et al., "Soil landscapes of the United States (SOLUS): Developing predictive soil property maps of the conterminous United States using hybrid training sets," *Soil Sci. Soc. Amer. J.*, vol. 88, no. 6, pp. 2046–2065, Nov. 2024, doi: 10.1002/saj2.20769.
[43] Q. Mu, M. Zhao, and S. W. Running, "Improvements to a MODIS global terrestrial evapotranspiration algorithm," *Remote Sens. Environ.*, vol. 115, no. 8, pp. 1781–1800, Aug. 2011, doi: 10.1016/j.rse.2011.02.019.
[44] A. AghaKouchak, "A baseline probabilistic drought forecasting framework using standardized soil moisture index: Application to the 2012 United States drought," *Hydrol. Earth Syst. Sci.*, vol. 18, no. 7, pp. 2485–2492, Jul. 2014, doi: 10.5194/hess-18-2485-2014.
[45] Z. Han et al., "Spatial-temporal dynamics of agricultural drought in the loess Plateau under a changing environment: Characteristics and potential influencing factors," *Agricult. Water Manage.*, vol. 244, Feb. 2021, Art. no. 106540, doi: 10.1016/j.agwat.2020.106540.
[46] L. Xu, N. Chen, Z. Chen, C. Zhang, and H. Yu, "Spatiotemporal forecasting in Earth system science: Methods, uncertainties, predictability and future directions," *Earth-Sci. Rev.*, vol. 222, Nov. 2021, Art. no. 103828, doi: 10.1016/j.earscirev.2021.103828.
[47] P. X. Flanagan, "Precipitation and temperature maxima: A study across the southern great plains winter wheat region," *Earth Interact.*, vol. 28, no. 1, Jan. 2024, Art. no. 230019, doi: 10.1175/ei-d-23-0019.1.
[48] Y. Wang et al., "Generation of root zone soil moisture from the integration of an all-weather satellite surface soil moisture estimates and an analytical model: A preliminary result in China," *J. Hydrol.*, vol. 644, Nov. 2024, Art. no. 132098, doi: 10.1016/j.jhydrol.2024.132098.
[49] F. Faridani, A. Farid, H. Ansari, and S. Manfreda, "A modified version of the SMAR model for estimating root-zone soil moisture from time-series of surface soil moisture," *Water SA*, vol. 43, no. 3, p. 492, Aug. 2017, doi: 10.4314/wsa.v43i3.14.
[50] K. E. Saxton and W. J. Rawls, "Soil water characteristic estimates by texture and organic matter for hydrologic solutions," *Soil Sci. Soc. Amer. J.*, vol. 70, no. 5, pp. 1569–1578, Sep. 2006, doi: 10.2136/sssaj2005.0117.
[51] G. K. Adamu and A. K. Aliyu, "Determination of the influence of texture and organic matter on soil water holding capacity in and around tomas irrigation scheme, dambatta local government Kano state," *Res. J. Environ. Earth Sci.*, vol. 4, no. 12, pp. 1038–1044, Dec. 2012.
[52] L. Tian, S. Yuan, and S. M. Quiring, "Evaluation of six indices for monitoring agricultural drought in the south-central United States," *Agricult. Forest Meteorol.*, vol. 249, pp. 107–119, Feb. 2018, doi: 10.1016/j.agrformet.2017.11.024.
[53] E. S. Krueger, T. E. Ochsner, and S. M. Quiring, "Development and evaluation of soil moisture-based indices for agricultural drought monitoring," *Agronomy J.*, vol. 111, no. 3, pp. 1392–1406, Mar. 2019, doi: 10.2134/agronj2018.09.0558.
[54] Z. Hao and A. AghaKouchak, "A nonparametric multivariate multi-index drought monitoring framework," *J. Hydrometeorol.*, vol. 15, no. 1, pp. 89–101, Feb. 2014, doi: 10.1175/jhm-d-12-0160.1.
[55] A. Mirdarsoltany, A. B. Dariane, and M. I. Borhan, "Comprehensive GIS-driven evaluation of drought severity and duration: Comparative assessment of parametric and non-parametric SPI methodologies," *Theor. Appl. Climatol.*, vol. 156, no. 4, p. 225, Apr. 2025, doi: 10.1007/s00704-025-05448-5.
[56] N. K. C. Twarakavi, M. Sakai, and J. Šimunek, "An objective analysis of the dynamic nature of field capacity," *Water Resour. Res.*, vol. 45, no. 10, p. 2009, Oct. 2009, doi: 10.1029/2009wr007944.

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